

# 2D ARRAY ASSIGNMENT USING NUMPY

Complete code for all the questions.

Student numbering uses 0-based indexing: Student 0 is the first row, Student 1 is the second row, and Student 2 is the third row.

```
import numpy as np

# 2D array of marks
# Student 0, Student 1, Student 2, Student 3, Student 4

marks = np.array([
    [80, 70, 90],
    [60, 75, 85],
    [90, 88, 95],
    [55, 65, 70],
    [78, 82, 80]
])

# i. Maximum marks
print("Maximum marks:", np.max(marks))

# ii. Minimum marks
print("Minimum marks:", np.min(marks))

# iii. Average marks
print("Average marks:", np.mean(marks))

# iv. Maximum marks subject-wise
print("Maximum marks subject-wise:", np.max(marks, axis=0))

# v. Average marks subject-wise
print("Average marks subject-wise:", np.mean(marks, axis=0))

# vi. Add 10 marks to students who scored less than 50 in Subject 1
for i in range(5):
    if marks[i][0] < 50:
        marks[i][0] = marks[i][0] + 10

print("Marks after adding 10:", marks)

# vii. Number of students who scored more than 80 in Subject 2
count = 0

for i in range(5):
    if marks[i][1] > 80:
        count = count + 1

print("Number of students:", count)

# viii. Minimum marks of Student 2
print("Minimum marks of Student 2:", np.min(marks[2]))
```

## Expected Output

```
Maximum marks: 95
Minimum marks: 55
Average marks: 77.53333333333333
Maximum marks subject-wise: [90 88 95]
Average marks subject-wise: [72.6 76.  84. ]
Marks after adding 10:
[[80 70 90]
 [60 75 85]
 [90 88 95]
 [55 65 70]
 [78 82 80]]
```

Number of students: 2  
Minimum marks of Student 2: 88